

T3 BSM UV terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from BSM UV terms. Empty cells denote no UV term with that field combination.

Field combinations 1 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	S_1	S_2	$F D$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\overline{N_{a_1, r_1}} N_{a_1, r_1} M_F$	$-\text{NewScalar1NewScalar1}^\dagger M_{S_1}^2$ $+\frac{1}{2}\text{lambdaS1Inv1NewScalar1}^2 \left(\text{NewScalar1}^\dagger \right)^2$	$-M_{S_1}^2 \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger$ $+\frac{1}{2}\text{loop-delta1Inv1NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger$ $+\frac{1}{2}\text{loop-delta1Inv1NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger$	$+i\overline{N_{a_1, r_1}} \gamma^{\mu 1} D_{\mu 1} \left(N_{a_1, r_1} \right)$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{1}{2} N_{r_1}^T C N_{r_1} M_F$	$-\text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger M_F^2$ $+\frac{1}{2}\text{loop-delta1Inv1NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger M_F^2$	$-M_{S_1}^2 \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger$ $+\frac{1}{2}\text{loop-delta1Inv1NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger$	$+\frac{1}{2} i N_{r_1}^T C \gamma^{\mu 1} D_{\mu 1} \left(N_{r_1} \right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{1}{2} N_{A_1, r_1}^T C N_{A_1, r_1} M_F$	$-\text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger M_F^2$ $+\frac{1}{2}\text{loop-delta1Inv1NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger M_F^2$	$-M_{S_1}^2 \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger$ $+\frac{1}{2}\text{loop-delta1Inv1NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger$	$+\frac{1}{2} i N_{A_1, r_1}^T C \gamma^{\mu 1} D_{\mu 1} \left(N_{A_1, r_1} \right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\overline{N_{a_1, r_1}} N_{a_1, r_1} M_F$	$-\text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger M_F^2$ $+\frac{1}{2}\text{loop-delta1Inv1NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger M_F^2$ $+\frac{1}{2}\text{loop-delta1Inv1NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger$	$+\frac{1}{2}\text{lambdaS2Inv1NewScalar2}^2 \left(\text{NewScalar2}^\dagger \right)^2$ $-\text{NewScalar2} M_{S_2}^2 \text{NewScalar2}^\dagger$	$+i\overline{N_{a_1, r_1}} \gamma^{\mu 1} D_{\mu 1} \left(N_{a_1, r_1} \right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\overline{N_{a_1, r_1}} N_{a_1, r_1} M_F$	$-\text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger M_F^2$ $+\frac{1}{2}\text{loop-delta1Inv1NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger M_F^2$ $+\frac{1}{2}\text{loop-delta1Inv1NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger$	$-M_{S_1}^2 \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger$ $+\frac{1}{2}\text{loop-delta1Inv1NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger$ $+\frac{1}{2}\text{loop-delta1Inv1NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger$	$+i\overline{N_{a_1, r_1}} \gamma^{\mu 1} D_{\mu 1} \left(N_{a_1, r_1} \right)$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	D	S_1	H	S_1	S_2	S_2	D	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}$	(NewScalar1) $^\dagger D_{\mu_1}$	(NewScalar1)	$+i\text{ambdaH1Inv1NewScalar1}H_{a_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2}\text{NewScalar1}^\dagger H_{a_1}^\dagger$	$+i\text{ambda12Inv1NewScalar1NewScalar2}_{a_2}\text{T3CrossStraleCGInv1}_{a_1,a_2}\text{NewScalar1}^\dagger\text{NewScalar2}_{a_1}^\dagger$	$+D_{\mu_1}$	(NewScalar2 A_1) $^\dagger D_{\mu_1}$	(NewScalar2 A_1)	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}$	(NewScalar1 a_1) $^\dagger D_{\mu_1}$	(NewScalar1 a_1)	$+i\text{ambdaH1Inv1NewScalar1}_{a_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2,a_2,a_2}H_{a_1}^\dagger\text{NewScalar1}_{a_2}^\dagger$	$+i\text{ambda12Inv1NewScalar1}_{a_2}\text{NewScalar2}_{a_2}\text{T3CrossStraleCGInv1}_{a_1,a_2,a_2,a_2}\text{NewScalar1}_{a_2}^\dagger\text{NewScalar2}_{a_1}^\dagger$	$+D_{\mu_1}$	(NewScalar2 a_1) $^\dagger D_{\mu_1}$	(NewScalar2 a_1)	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}$	(NewScalar1 a_1) $^\dagger D_{\mu_1}$	(NewScalar1 a_1)	$+i\text{ambdaH1Inv1NewScalar1}_{a_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2,a_2,a_2}H_{a_1}^\dagger\text{NewScalar1}_{a_2}^\dagger$	$+i\text{ambda12Inv1NewScalar1}_{a_2}\text{NewScalar2}_{a_2}\text{T3CrossStraleCGInv1}_{a_1,a_2,a_2,a_2}\text{NewScalar1}_{a_2}^\dagger\text{NewScalar2}_{a_1}^\dagger$	$+D_{\mu_1}$	(NewScalar2 a_1) $^\dagger D_{\mu_1}$	(NewScalar2 a_1)	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}$	(NewScalar1 A_1) $^\dagger D_{\mu_1}$	(NewScalar1 A_1)	$+i\frac{1}{2}\text{ambdaH1Inv1NewScalar1}_{a_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2,A_1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$					
											$+i\frac{1}{2}\text{ambdaH1Inv1NewScalar1}_{a_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2,A_1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$		$+i\text{ambda12Inv1NewScalar1NewScalar2NewScalar1}_{a_2}\text{T3CrossStraleCGInv1}_{a_1,a_2,a_2,a_2}\text{NewScalar1}_{a_2}^\dagger\text{NewScalar2}_{a_1}^\dagger$	$+D_{\mu_1}$	(NewScalar2) $^\dagger D_{\mu_1}$	(NewScalar2)
											$+i\frac{1}{2}\text{ambdaH1Inv2NewScalar1}_{a_2}\text{T3HiggsPortal1CGInv2}_{a_1,a_2,A_1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$					
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}$	(NewScalar1 A_1) $^\dagger D_{\mu_1}$	(NewScalar1 A_1)	$+i\frac{1}{2}\text{ambdaH1Inv1NewScalar1}_{a_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2,A_1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$					
											$+i\frac{1}{2}\text{ambdaH1Inv1NewScalar1}_{a_2}\text{T3HiggsPortal1CGInv1}_{a_1,a_2,A_1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$		$+i\text{ambda12Inv1NewScalar1}_{a_2}\text{NewScalar2}_{a_2}\text{T3CrossStraleCGInv1}_{a_1,a_2,a_2,a_2}\text{NewScalar1}_{a_2}^\dagger\text{NewScalar2}_{a_1}^\dagger$	$+D_{\mu_1}$	(NewScalar2 A_1) $^\dagger D_{\mu_1}$	(NewScalar2 A_1)
											$+i\frac{1}{2}\text{ambdaH1Inv2NewScalar1}_{a_2}\text{T3HiggsPortal1CGInv2}_{a_1,a_2,A_1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$		$+i\text{ambda12Inv1NewScalar1}_{a_2}\text{NewScalar2}_{a_2}\text{T3CrossStraleCGInv1}_{a_1,a_2,a_2,a_2}\text{NewScalar1}_{a_2}^\dagger\text{NewScalar2}_{a_1}^\dagger$			

Field combinations 3 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 H$	$S_1 F \ell$	$S_1 S_2 H$	$S_2 F \ell$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{2}\text{Im bda}H_2\text{inv}H_{A_2}\text{New Scalar}f_{2A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{1,1}_{A_1,A_2,A_3}H_{A_1}{}^{\dagger}\text{New Scalar}f_{2A_1}{}^{\dagger}$ $+\frac{1}{2}\text{Im bda}H_2\text{inv}H_{A_2}\text{New Scalar}f_{2A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{1,1}_{A_1,A_2,A_3}H_{A_1}{}^{\dagger}\text{New Scalar}f_{2A_1}{}^{\dagger}$ $+\frac{1}{2}\text{Im bda}H_2\text{inv}H_{A_2}\text{New Scalar}f_{2A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{2,2}_{A_1,A_2,A_3}H_{A_1}{}^{\dagger}\text{New Scalar}f_{2A_1}{}^{\dagger}$ $+\frac{1}{2}\text{Im bda}H_2\text{inv}H_{A_2}\text{New Scalar}f_{2A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{2,2}_{A_1,A_2,A_3}H_{A_1}{}^{\dagger}\text{New Scalar}f_{2A_1}{}^{\dagger}$	$+\text{New Scalar}i\overline{\ell_{A_1,1}}CPR\overline{N_{A_2,r_1}}^Ty_{1,1},r_1\epsilon_{A_1,a_2}$ $+\overline{y_{1,1},r_1}\epsilon_{A_1,i_1}{}^TCPLN_{A_2,r_1}\overline{\epsilon_{A_1,a_2}}\text{New Scalar}1_{A_1}{}^{\dagger}$	$+\overline{\lambda_{T3}}\text{New Scalar}2_{A_1}\epsilon_{a_2,a_3}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_3}\text{New Scalar}1_{A_1}{}^{\dagger}H_{A_2}{}^{\dagger}H_{A_2}$ $+\text{New Scalar}1_{A_2}H_{A_2}\overline{\lambda_{T3}}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_3}\epsilon_{a_1,a_2}\text{New Scalar}2_{A_1}{}^{\dagger}$	$+\overline{y_{2,1},r_1}\overline{N_{A_1,r_1}}PL\ell_{A_2,i_1}\text{New Scalar}2_{A_1}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_2}$ $+\overline{\ell_{A_1,i_1}}PRN_{A_2,r_1}y_{2,1},r_1\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_2}\text{New Scalar}2_{A_1}{}^{\dagger}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\text{Im bda}H_2\text{inv}H_{A_2}\text{New Scalar}2_{A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{1,1}_{A_1,A_2,A_3}H_{A_1}{}^{\dagger}\text{New Scalar}2_{A_2}{}^{\dagger}$ $+\text{Im bda}H_2\text{inv}H_{A_2}\text{New Scalar}2_{A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{2,2}_{A_1,A_2,A_3}H_{A_1}{}^{\dagger}\text{New Scalar}2_{A_2}{}^{\dagger}$	$+\overline{\ell_{A_1,1}}PRN_{r_1}\text{New Scalar}1_{A_1}y_{1,i_1},r_1$ $+\overline{y_{1,1},r_1}\overline{N_{r_1}}{}^TCPL\ell_{A_1,i_1}\text{New Scalar}1_{A_1}{}^{\dagger}$	$+\overline{\lambda_{T3}}\text{New Scalar}2_{A_1}\epsilon_{a_2,a_3}H_{A_1}{}^{\dagger}H_{A_2}{}^{\dagger}\text{New Scalar}1_{A_3}$ $+H_{A_1}H_{A_2}\text{New Scalar}1_{A_3}\overline{\lambda_{T3}}\epsilon_{a_2,a_3}\text{New Scalar}2_{A_1}{}^{\dagger}$	$+\overline{y_{2,1},r_1}\overline{N_{r_1}}{}^TCPL\ell_{A_1,i_1}\text{New Scalar}2_{A_2}\overline{\epsilon_{A_1,a_2}}$ $+\overline{\ell_{A_2,i_1}}PRN_{r_1}y_{2,1},r_1\epsilon_{A_2,a_1}\text{New Scalar}2_{A_1}{}^{\dagger}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\text{Im bda}H_2\text{inv}H_{A_2}\text{New Scalar}2_{A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{1,1}_{A_1,A_2,A_3}H_{A_1}{}^{\dagger}\text{New Scalar}2_{A_2}{}^{\dagger}$ $+\text{Im bda}H_2\text{inv}H_{A_2}\text{New Scalar}2_{A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{2,2}_{A_1,A_2,A_3}H_{A_1}{}^{\dagger}\text{New Scalar}2_{A_2}{}^{\dagger}$	$+\overline{\ell_{A_1,1}}PRN_{A_1,r_1}\text{New Scalar}1_{A_2}y_{1,i_1},r_1\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_2}$ $+\overline{y_{1,1},r_1}\overline{N_{A_1,r_1}}{}^TCPL\ell_{A_2,i_1}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_2}\text{New Scalar}1_{A_1}{}^{\dagger}$	$+\overline{\lambda_{T3}}\text{New Scalar}2_{A_1}\epsilon_{a_2,a_3}H_{A_1}{}^{\dagger}H_{A_2}{}^{\dagger}\text{New Scalar}1_{A_3}$ $+H_{A_1}H_{A_2}\text{New Scalar}1_{A_3}\overline{\lambda_{T3}}\epsilon_{a_2,a_3}\text{New Scalar}2_{A_1}{}^{\dagger}$	$+\overline{y_{2,1},r_1}\overline{N_{A_1,r_1}}{}^TCPL\ell_{A_3,i_1}\text{New Scalar}2_{A_2}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_3}\epsilon_{a_1,a_2}$ $+\overline{\ell_{A_3,i_1}}PRN_{A_1,r_1}y_{2,1},r_1\epsilon_{a_2,a_3}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_3}\text{New Scalar}2_{A_1}{}^{\dagger}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\text{Im bda}H_2\text{inv}1\text{New Scalar}2_{A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{1,1}_{A_1,a_2}\text{New Scalar}2_{A_2}{}^{\dagger}H_{A_1}{}^{\dagger}$ $+\overline{y_{A_1,r_1}}{}^TCPLN_{A_1,r_1}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_2}\epsilon_{a_1,a_2}\text{New Scalar}1_{A_1}{}^{\dagger}$	$+\overline{\ell_{A_3,i_1}}CPH\overline{N_{A_1,r_1}}{}^T\text{New Scalar}1_{A_1}y_{1,i_1},r_1\epsilon_{a_1,a_2}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_2}$ $+\overline{y_{A_1,r_1}}{}^TCPLN_{A_1,r_1}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_2}\epsilon_{a_1,a_2}\text{New Scalar}1_{A_1}{}^{\dagger}$	$+H_{A_2}H_{A_2}\text{New Scalar}1_{A_1}\overline{\lambda_{T3}}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_2}\epsilon_{a_1,a_2}\text{New Scalar}2_{A_1}{}^{\dagger}$ $+\text{New Scalar}2_{A_1}\overline{\lambda_{T3}}\epsilon_{a_3,a_2}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_3}H_{A_1}{}^{\dagger}H_{A_2}{}^{\dagger}\text{New Scalar}1_{A_1}{}^{\dagger}$	$+\text{New Scalar}2_{A_2}y_{2,i_1},r_1\overline{N_{A_1,r_1}}PL\ell_{A_1,i_1}$ $+\overline{\ell_{A_1,i_1}}PRN_{A_1,r_1}y_{2,i_1},r_1\text{New Scalar}2_{A_1}{}^{\dagger}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{2}\text{Im bda}H_2\text{inv}H_{A_2}\text{New Scalar}f_{2A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{1,1}_{A_1,A_2,A_3}H_{A_1}{}^{\dagger}\text{New Scalar}f_{2A_1}{}^{\dagger}$ $+\frac{1}{2}\text{Im bda}H_2\text{inv}H_{A_2}\text{New Scalar}f_{2A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{1,1}_{A_1,A_2,A_3}H_{A_1}{}^{\dagger}\text{New Scalar}f_{2A_1}{}^{\dagger}$ $+\frac{1}{2}\text{Im bda}H_2\text{inv}H_{A_2}\text{New Scalar}f_{2A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{2,2}_{A_1,A_2,A_3}H_{A_1}{}^{\dagger}\text{New Scalar}f_{2A_1}{}^{\dagger}$ $+\frac{1}{2}\text{Im bda}H_2\text{inv}H_{A_2}\text{New Scalar}f_{2A_2}T_3H_{gg}\text{Porta}t2CG\text{Inv}^{2,2}_{A_1,A_2,A_3}H_{A_1}{}^{\dagger}\text{New Scalar}f_{2A_1}{}^{\dagger}$	$+\overline{\ell_{A_3,i_1}}CPH\overline{N_{A_1,r_1}}{}^T\text{New Scalar}1_{A_1}y_{1,i_1},r_1\epsilon_{a_1,a_2}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_2}$ $+\overline{y_{A_1,r_1}}{}^TCPLN_{A_1,r_1}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_2}\epsilon_{a_1,a_2}\text{New Scalar}1_{A_1}{}^{\dagger}$	$+\overline{\lambda_{A_1}}\text{New Scalar}2_{A_2}\epsilon_{a_2,a_3}\text{gen}(\text{SU}(2L_1, A_1), A_3)\text{gen}(\text{SU}(2L_1, \text{fund}))_{A_1,a_1,a_3}H_{A_2}{}^{\dagger}H_{A_2}\text{New Scalar}1_{A_1}{}^{\dagger}$ $-H_{A_2}\text{New Scalar}1_{A_3}\overline{\lambda_{A_1}}\text{gen}(\text{SU}(2L_1, A_1), A_3)\epsilon_{a_2,a_3}\text{gen}(\text{SU}(2L_1, \text{fund}))_{A_1,a_1,a_3}\epsilon_{a_1,a_2}\text{New Scalar}2_{A_1}{}^{\dagger}$	$+\overline{y_{2,1},r_1}\overline{N_{A_1,r_1}}PL\ell_{A_2,i_1}\text{New Scalar}2_{A_1}\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_2}$ $+\overline{\ell_{A_1,i_1}}PRN_{A_2,r_1}y_{2,i_1},r_1\text{gen}(\text{SU}(2, \text{fund}))_{A_1,a_1,a_2}\text{New Scalar}2_{A_1}{}^{\dagger}$